Experimentation in the Age of Generative AI: Evidence from ChatGPT

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Extended Abstract

The advent of Generative AI (GenAI), exemplified by large language models such as ChatGPT, is transforming how startups are conceived, tested, and built. Tools that were initially viewed as productivity aids for individual founders now permeate core entrepreneurial processes, from ideation and product design to customer interaction and fundraising. As GenAI diffuses rapidly across the startup ecosystem, it becomes essential to understand not only whether it makes entrepreneurs "faster and cheaper," but also how it reshapes the fundamental ways in which they search, learn, and make strategic decisions.

Beyond automating routine tasks, GenAI can generate ideas, write code, simulate customer interactions, and synthesize market information, potentially reshaping a core entrepreneurial activity: *experimentation*. In many settings, experimentation is formalized through the "Lean Startup" methodology (Contigiani and Levinthal, 2019; Wang et al., 2025), which emphasizes iterative cycles of hypothesis formulation, testing minimum viable products (MVP) in the real world, and rapid adaptation to feedack. A growing literature documents that rigorous experimentation, in its various facets, can substantially improve startup performance and growth (Koning et al., 2022; Camuffo et al, 2024). In this paper, we explore how the role of experimentation changes in the age of GenAI. More precisely, we ask the following question: how does the emergence of GenAI affect startups' use of experimentation?

We build on research in entrepreneurship and organizational learning to theorize two competing mechanisms, which we label the Substitution Hypothesis and the Productivity Hypothesis. The Substitution Hypothesis posits that GenAI may crowd out traditional experimentation. By providing low-cost access to a variety of information – including synthetic customer feedback, scenario analysis, and expert-like advice -- GenAI can increase the returns to non-experimental learning. In other words, in the age of GenAI, firms can learn "on the screen" rather than "in the field." If founders perceive GenAI insights as sufficiently informative, they may reduce costly MVP tests, customer interviews, and A/B experiments. In contrast, the Productivity Hypothesis views GenAI as a complement that changes the efficiency of experimentation. By lowering the cost of generating prototypes, user interfaces, marketing materials, or analytical scripts, GenAI may make each testing cycle cheaper and faster, enabling ventures to learn more per unit of experimental effort and potentially reducing how much effort is allocated to experimentation-related activities. This perspective complements recent evidence that AI investments can enhance product innovation when paired with Lean Startup practices such as prototyping and controlled experimentation (Wang et al., 2025).

Both hypotheses point to a negative relationship between GenAI and experimentation. As GenAI becomes available, new ventures may rely less on human labor for experimentation, either because the benefit of additional experimentation is lower (substitution) or because the productivity of each experiment is higher (productivity).

To examine this hypothesis, we exploit the November 2022 public release of ChatGPT as a sudden, widely diffused technological shock. Recent research uses the same shock to examine the effect of GenAI on firm formation (Marchesi et al., 2025). We extend this line of work by shifting the focus from entry to within-firm learning, studying how GenAI affects

experimentation capabilities inside existing ventures. We assemble a novel panel of more than 30,000 U.S. software ventures founded between 2018 and 2022 and follow their behavior until mid 2025.

Our empirical strategy requires identifying a treatment group. As this shock, in principle, affects everyone, we proceed by building a measure of exposure. We construct an ex-ante measure of GenAI exposure based on founders' backgrounds. Using detailed information on top management team (TMT) members, we identify ventures whose founders had pre-existing Deep Learning or closely related AI/ML skills prior to the ChatGPT launch. We assume that these ventures are more "ready" to understand, adopt, and integrate GenAI tools into their workflow, once they become widely available.

We plan to measure experimentation in several ways. In the current version of the paper, we measure experimentation intensity using job descriptions collected from Revelio Labs. We collect and examine job descriptions to identify roles that explicitly involve experimentation-related tasks—such as product experimentation, A/B testing, rapid prototyping, user research, and data-driven product analytics—and we track how the prevalence of these roles evolves over time. Following the literature, we focus on two well-known types of experimentation: prototyping (experimentation of type 1, involving MVP development and qualitative product tests) and A/B testing (experimentation of type 2, involving formal, data-driven online experiments). Distinguishing between these two forms allows us to study whether GenAI displaces early-stage, design-oriented experimentation more than later-stage, analytics-heavy experimentation.

We employ a difference-in-differences (DiD) design that compares changes in hiring between GenAI-exposed and non-exposed ventures before and after the ChatGPT shock. The

main specification includes venture and time fixed effects, absorbing time-invariant venture characteristics and common macro trends. An event-study framework tests the parallel trends assumption and allows us to visualize dynamic treatment effects around the November 2022 breakpoint, highlighting both immediate and lagged responses to the shock. We further probe robustness using alternative exposure definitions, sample restrictions, and specifications that account for venture age, funding stage, and sectoral heterogeneity.

Before analyzing our hypothesis, we explore the effect on firm size. We find that GenAI-exposed ventures expand hiring following the ChatGPT launch. Relative to otherwise similar ventures without GenAI-ready founders, these firms display employment growth in the post-shock period, suggesting that GenAI adoption is associated with organizational growth rather than downsizing or automation-induced contraction. This result is consistent with recent evidence that AI tools like chatbots correlate with increased startup employment (Impink et al, 2025), particularly when ventures possess complementary technological capabilities and are operationally ready to scale, and it aligns with work documenting that ChatGPT's launch reshaped labor markets and induced strategic repositioning among freelancers (Yiu et al., 2024).

As we proceed to our main analysis, we find evidence that experimentation-related employment declines among GenAI-exposed startups, consistent with our hypotheses. The ratio of experimentation-related positions to total jobs decreases post-ChatGPT, indicating that experimentation roles become relatively less central in the hiring portfolios of GenAI-ready firms. This result is particularly strong for experimentation of type 1 (prototyping and MVP development), but it is also visible—albeit more modestly—for type 2 (A/B testing and formal experimentation). This pattern is robust across a range of checks. Event-study estimates show no meaningful pre-trends in experimentation-related hiring, lending credibility to the DiD design.

Results hold when we vary the definition of GenAI exposure (e.g., broad vs. narrow sets of AI-related skills), exclude firms founded very close to the shock, or control flexibly for venture age, funding history, and sector-specific time trends. Taken together, the findings suggest that while experimentation remains important for startup performance and learning, GenAI-ready ventures grow overall but lean relatively less on experimentation specialists, potentially because GenAI tools substitute for some tasks previously performed by these roles.

We are currently extending the analysis to shed light on the underlying mechanisms, in particular trying to distinguish between substitution and productivity effects. On the quantitative side, we are examining heterogeneity across product domains, business models, and founder backgrounds to identify where experimentation employment contracts the most. We are also broadening the set of outcome variables beyond hiring. Specifically, we will study the use of experimentation-related tools—such as A/B testing and product-analytics platforms—by tracking changes in technology stacks using BuiltWith data (Stroube and Dushnitsky, 2023). Further, we will examine innovation outputs by linking ventures to trademark filings from the USPTO, and access to financing by studying patterns in seed and venture capital funding using data from Crunchbase and PitchBook. On the qualitative side, we are conducting semi-structured interviews and case studies with founders in GenAI-exposed ventures to understand how they integrate tools like ChatGPT and related APIs into their product development workflows, when they choose to replace real-world tests with digital simulations, and how they reconfigure their teams and experimentation practices as a result.

We believe this study makes three contributions. First, it adds to the emerging literature on GenAI in organizations by providing one of the first large-scale, quasi-experimental analyses of how GenAI affects entrepreneurial search and learning processes at the firm level. Whereas

existing work has primarily examined how ChatGPT's commercialization spurred new firm creation in AI-compatible industries or how specific AI tools affect labor outcomes, we show how a technological shock reshapes hiring, technology adoption, and capability-building around experimentation. Second, we contribute to entrepreneurship and Lean Startup research by documenting that GenAI can be viewed as an alternative "learning process" and reconfigure the role of real-world testing. Even as overall hiring grows, GenAI-ready ventures devote a smaller share of their human capital to experimentation roles, suggesting that AI and Lean Startup methods can be either complements or partial substitutes depending on how they are combined. Third, we enrich research on organizational learning and technological change by documenting how prior human capital—founders' Deep Learning skills—conditions firms' responses to a new general-purpose technology, influencing both the scale of hiring and the composition of capabilities they build.

More broadly, this work speaks to ongoing debates about whether GenAI will encourage "armchair entrepreneurship" built primarily on digital simulations or whether it will foster ever more empirical, experiment-driven company building. So far, our evidence tilts toward a nuanced version of the former: when used by GenAI-ready founders, ChatGPT and related tools appear to enable startups to grow while relying relatively less on labor-intensive experimentation roles, even as experimentation technologies and Lean practices likely remain crucial for performance. This shift raises important questions about the quality of learning under GenAI, the kinds of ideas that get selected when experimentation is partially digitized, and the long-run implications for innovation, funding dynamics, and labor in entrepreneurial ecosystems.

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